고려대학교 약학대학
KOREA UNIVERSITY COLLEGE OF PHARMACY

KU-The Future
110 years and beyond

Microbiology

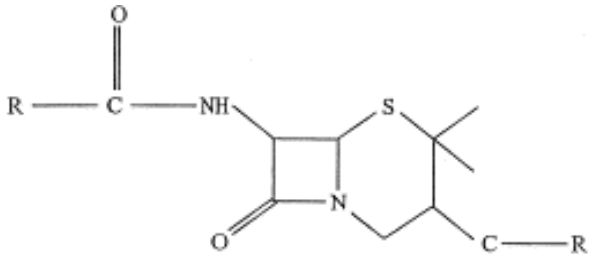


Fall semester, 2026

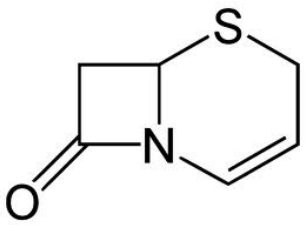
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❖ Cephems

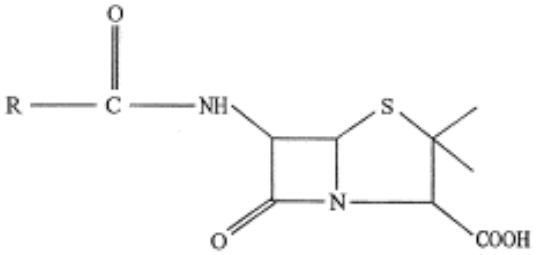
- cephalosporins
- cephamycins



Cephalosporin Nucleus



Cephem nucleus



Penicillin Nucleus

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❖ Cephalosporins

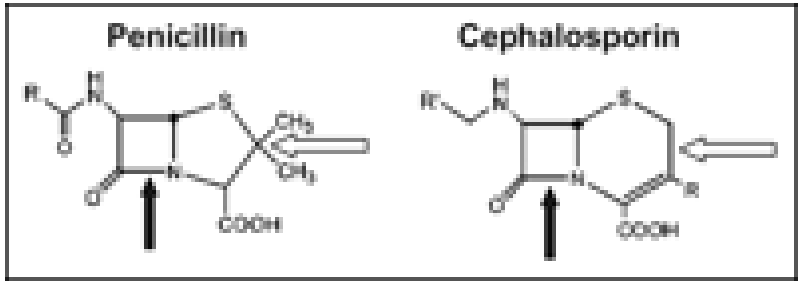
· a class of β-lactam antibiotics; containing a β-lactam ring (structurally similar to penicillins) → cephem

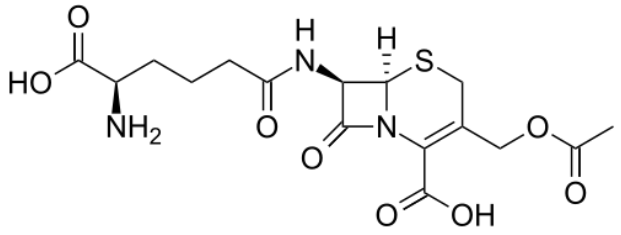
· natural cephalosporins (e.g. cephalosporin P, N and C)
: derived from the fungus *Acremonium* (*Cephalosporium*)

· grouped into "generations (1 to 5)" based on spectrum of activity

· <u>several things in common</u>:

- having some cross-allergenicity with penicillins (<10%)
- in general, more resistant to β-lactamase than penicillins are
- none of them have useful activity against *enterococci* (except some)





Cephalosporin C

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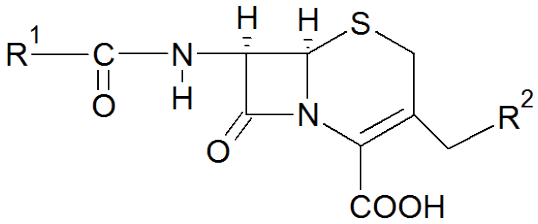
▶ First-generation cephalosporins

· ex) cephalothin, cephaloridine, cefazolin, cephalexin, etc.

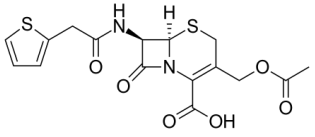
· the most commonly used class of antibiotics in the hospital
: used immediately prior to surgery to prevent surgical site infections

– why? spectrum of activity, inexpensive cost, low incidence of adverse effects, etc.

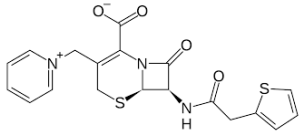
– spectrum of activity
· G (+), including penicillinase-producing MSSA (PRSA)
· some G (–) (e.g., some enteric GNRs)
· poor: MRSA, *Pseudomonas*, *enterococci*, etc.



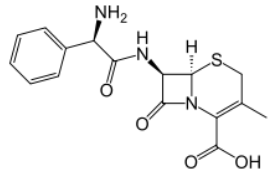
Core structure of cephalosporins
Basic structure (1st)



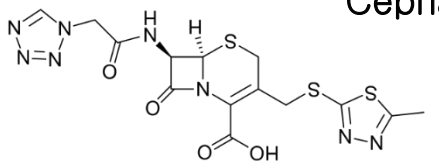
Cephalothin



Cephaloridine



Cephalexin

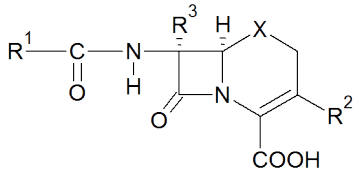


Cefazolin

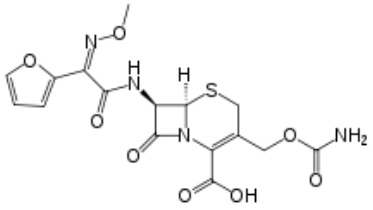
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▶ Second-generation cephalosporins

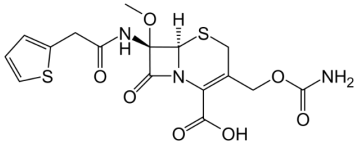
- ex) cefamandole, cefuroxime, cefonicid, **cefoxitin, cefotetan**, cefaclor, cefprozil, etc.
- spectrum of activity:
  - compared with 1st generation cephalosporins:
    - somewhat weaker G (+) activity (less than 1st)
    - having better G (−) bacteria activity (greater than 1st)
    - more stable against G (−) β-lactamases
      : active against some enteric GNRs
  - active against *Haemophilus influenzae* and *Neisseria gonorrheae*
  - poor: *Pseudomonas*, *enterococci*



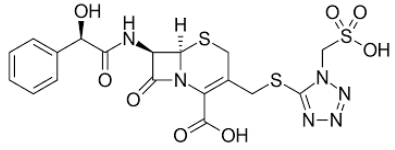
Basic structure (2nd)



Cefuroxime



Cefoxitin

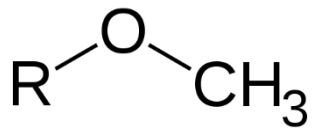


Cefonicid

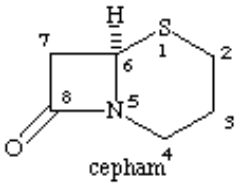
Second-generation cephalosporins

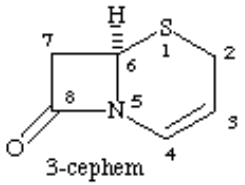
· some of them (e.g., **cefoxitin**, **cefotetan**) are called **cephamycins**

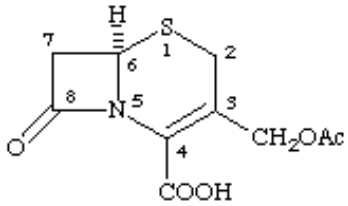
- containing a methoxy group at the 7-alpha position
- methoxy: functional group consisting of a methyl group bound to oxygen (R-O-CH3)



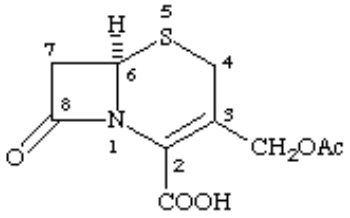
Methoxy group


cepham

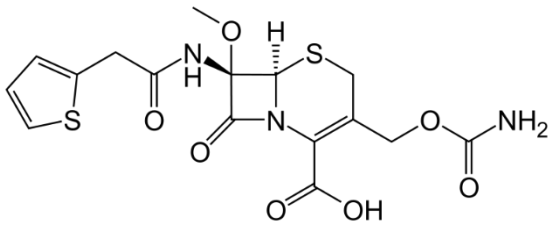

3-cephem

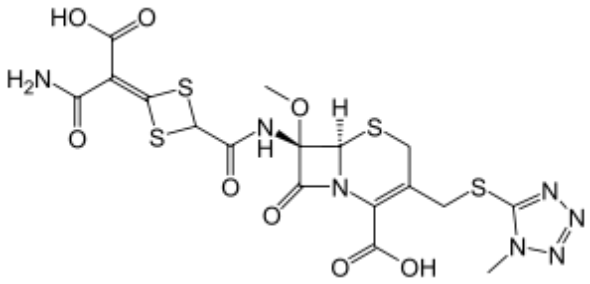


cephalosporanic acid



(6*R*)-3-(acetoxymethyl)-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid

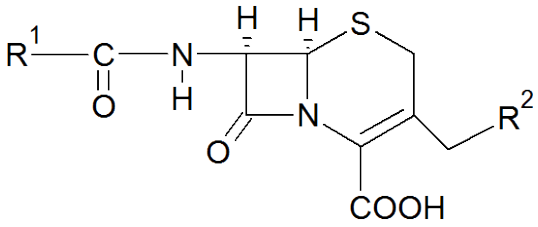

Cefoxitin


Cefotetan

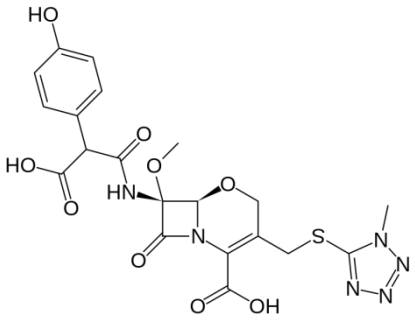
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▷ Oxacephems

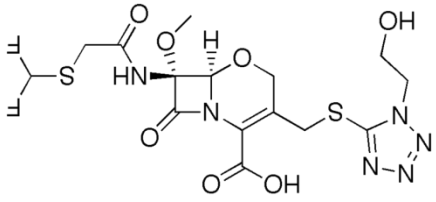
- molecules similar to cephems (cephalosporins and cephamycines)
- however, have oxygen substituted for the sulfur
- have a methoxy group at the 7-alpha position
- have similar antibacterial activity to the 3rd cephalosporins
- ex) **moxalactam** (latamoxef; currently not used), **flomoxef**



Core structure of cephalosporins
Basic structure (1st)



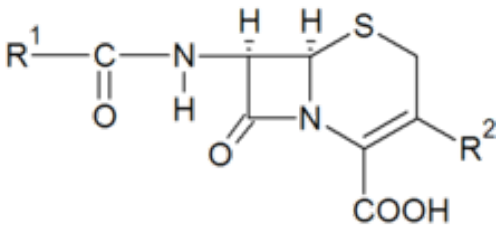
Moxalactam



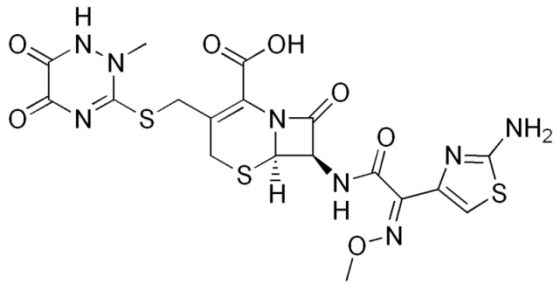
Flomoxef

▶ Third-generation cephalosporins

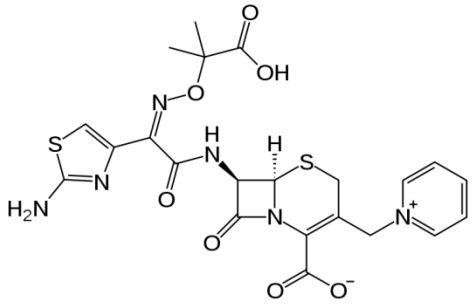
- cefotaxime, ceftriaxone, ceftazidime, cefdinir, etc.
- spectrum of activity:
  - G (+): some members have decreased activity
  - G (−): having greater G (−) activity than the 1st and 2nd generation cephalosporins → wide-spectrum
  - active against *Pseudomonas* (ceftazidime only) and enteric GNRs
  - poor: *Pseudomonas* and *enterococci*
- cefotaxime, ceftriaxone and ceftazidime cross the blood-brain barrier → useful for the treatment of CNS infections



Basic structure (3rd)



Ceftriaxone

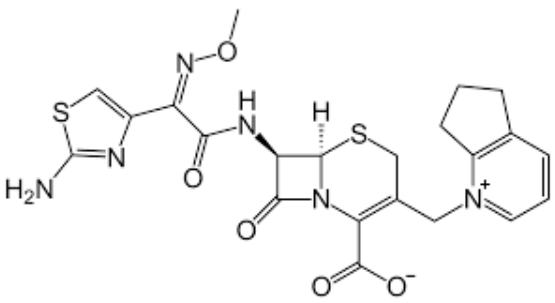


Ceftazidime

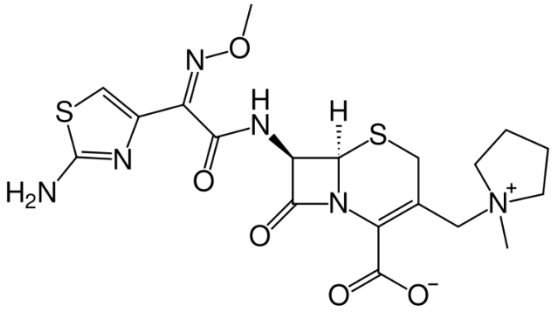
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▶ Fourth-generation cephalosporins

- cefpirome, cefepime, etc.
- spectrum of activity:
  - broad spectrum against G (+) and G (−) bacteria
    : better G (+) and G (−) activity
  - active against *Pseudomonas* and enteric GNRs
  - poor: *enterococci*



Cefpirome

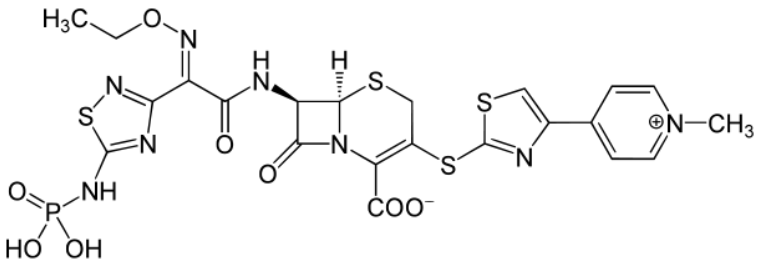


Cefepime

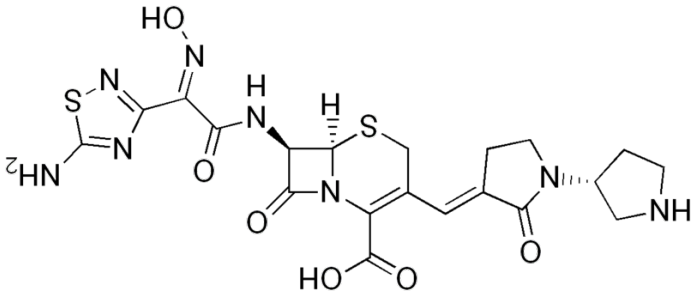
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▶ Fifth-generation cephalosporins

- ceftaroline, ceftobiprole
- only two cephalosporins in this group
  : recently developed
- spectrum of activity:
  - active against PRSA, MRSA, *Streptococci*, enteric GRNs, some *enterococci*
  - ceftaroline: not active against *Pseudomonas*
  - ceftobiprole: powerful antipseudomonal activity



Ceftaroline



Ceftobiprole